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Exam

One of the ten weekly exercises is drawn at random, and you talk through your own notebook for it. That is the whole exam.

Bring all ten, on a laptop or printed. An exercise you cannot produce is an exercise you did not do, and you will be sitting in front of it with nothing to say.

What the discussion asks

It is a conversation about work you have already done, not an examination of theory in the abstract. Expect to be asked:

- what you actually measured, and why that quantity rather than another;
- which decision in the notebook you were least sure of, and what you did about it;
- what result would have told you the whole thing was wrong;
- why the method you used was the right one for that problem — and what you would have reached for if the data had been different.

There are no marks for the score you obtained. There are marks for knowing why you obtained it. Two students with different numbers can both do well; a student with an excellent number who cannot say where it came from cannot.

That is the same standard the exercises are written to, and it is deliberate: the difficult skill in this subject is not producing a model that appears to work, but telling that apart from one that generalises.

Why the theory still matters

Nothing is asked in the abstract, but the questions above cannot be answered without it. “Why that metric” is a question about section 6 of lesson 4. “What would have told you it was wrong” is lesson 5. The handouts carry the derivations precisely so that the *why* is available to you when you are asked for it.

The practical consequence: re-read your own reasoning before the day rather than rewriting the code. The decisions live in the markdown cells of your notebooks, and those are what the discussion is about.

How to prepare

1. **Find all ten notebooks** and check each one runs. Do this in the week before, not on the day.
 2. **Read your own markdown cells.** Where a cell says what you did but not why, that is the question you will be asked.
 3. **Take the quizzes**, one per lesson, in `Lessons/NN_topic/Quizzes/`. They are written to require reasoning about a derivation rather than recalling a fact, which is the shape of the *why* questions.
 4. For the exercise you found hardest, work out in advance what you would do differently. Saying so is worth more than defending a choice you no longer believe.
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Academic integrity

You may use libraries freely unless an exercise asks for a from-scratch implementation, and you may use AI assistants to understand material or debug code. But **you must be able to explain every line you wrote**. Work you cannot explain is treated as work you did not write — and since the exam draws one of the ten for you to discuss, that is not a threat but a description of how the exercises are assessed.

Cite any source you drew on: datasets, papers, blog posts, code. A short “Sources” section at the end of a notebook is enough.